

APPENDIX B

STATISTICAL TABLES

Table B.1 Quantiles for the shortest CI for σ

Table B.2 Kolmogorov distribution

Table B.1 Quantiles of the chi-square distribution for determining the shortest CI for σ .
Relative gain in length can be found in the “%” column

ν	$\alpha = 0.10$				$\alpha = 0.05$			
	α_1	$\chi^2_{\alpha_1, \nu}$	$\chi^2_{\alpha_2, \nu}$	%	α_1	$\chi^2_{\alpha_1, \nu}$	$\chi^2_{\alpha_2, \nu}$	%
1	0.09998	0.016	18.189	24.92	0.04999	0.004	19.511	24.99
2	0.09988	0.211	18.056	49.02	0.04998	0.103	21.640	49.48
3	0.09948	0.582	17.647	61.21	0.04988	0.351	20.726	61.82
4	0.09882	1.056	18.100	68.50	0.04969	0.708	21.047	69.10
5	0.09800	1.594	18.907	73.41	0.04943	1.139	21.806	73.94
6	0.09708	2.175	19.871	76.95	0.04911	1.623	22.736	77.41
7	0.09612	2.788	20.927	79.65	0.04876	2.147	23.791	80.04
8	0.09516	3.426	22.041	81.77	0.04839	2.703	24.910	82.11
9	0.09420	4.084	23.182	83.48	0.04802	3.284	26.083	83.78
10	0.09328	4.758	24.352	84.90	0.04764	3.886	27.270	85.16
11	0.09238	5.447	25.530	86.09	0.04726	4.505	28.472	86.32
12	0.09152	6.147	26.719	87.11	0.04689	5.141	29.689	87.31
13	0.09070	6.858	27.915	87.98	0.04653	5.790	30.915	88.16
14	0.08990	7.579	29.109	88.75	0.04619	6.451	32.153	88.91
15	0.08916	8.308	30.313	89.42	0.04585	7.123	33.386	89.56
16	0.08844	9.045	31.514	90.02	0.04552	7.804	34.619	90.14
17	0.08774	9.788	32.711	90.55	0.04521	8.495	35.859	90.67
18	0.08710	10.539	33.916	91.03	0.04490	9.193	37.091	91.13
19	0.08648	11.295	35.117	91.46	0.04461	9.899	38.328	91.56
20	0.08588	12.056	36.315	91.85	0.04433	10.612	39.563	91.94
21	0.08530	12.823	37.510	92.21	0.04405	11.331	40.791	92.29
22	0.08476	13.595	38.708	92.54	0.04379	12.056	42.024	92.61
23	0.08424	14.371	39.903	92.84	0.04354	12.787	43.254	92.91
24	0.08374	15.151	41.095	93.12	0.04329	13.523	44.478	93.18
25	0.08326	15.935	42.286	93.37	0.04306	14.264	45.706	93.43
26	0.08278	16.723	43.471	93.61	0.04283	15.009	46.928	93.67
27	0.08234	17.514	44.659	93.83	0.04261	15.759	48.149	93.89
28	0.08192	18.310	45.846	94.04	0.04240	16.513	49.369	94.09
29	0.08150	19.108	47.027	94.23	0.04219	17.271	50.583	94.28
30	0.08110	19.909	48.208	94.41	0.04199	18.032	51.797	94.46
40	0.07780	28.063	59.927	95.75	0.04031	25.823	63.834	95.78
50	0.07536	36.418	71.499	96.57	0.03903	33.855	75.694	96.59
60	0.07346	44.918	82.949	97.13	0.03803	42.064	87.418	97.14
80	0.07066	62.232	105.571	97.83	0.03653	58.859	110.535	97.84
100	0.06866	79.847	127.915	98.25	0.03544	76.015	133.321	98.26

Table B.2 Tail probabilities $1 - Q(z)$ of the Kolmogorov distribution

[illegible]